

Dexterous Robotic Hand Grasp Modeling Using Piecewise Linear Dynamic Model

Wei Xiao¹, Fuchun Sun¹, Huaping Liu¹, Heyu Liu¹ and Chao He²

Abstract—Learning from sensor data is important in many robotic research areas, such as dexterous robotic hand grasping. In this paper, a piecewise linear dynamic model is proposed for analyzing robotic hand grasp. The combination of linear dynamic model and the switched systems can achieve better results in grasp learning due to its advantage of modeling multi-phase grasping process. To the best knowledge of the authors, this is the first time for piecewise linear dynamic model to be incorporated into the framework of modeling robotic hand grasp process. The performance of the proposed model is evaluated on our experimental system and shows promising results.

I. INTRODUCTION

“Anaxagoras says that because of having hands, man grew the most intelligent among animals. (I think) it is correct to say that because of his intelligence he has hands”. In one of his books on nature sciences [1], the Greek philosopher Aristotle (384-322 BC) argued with his late colleague Anaxagoras (500?-428 BC) over whether it was because humans had dexterous hands that they became intelligent, or the other way around. Anaxagoras’ intuition has been later on confirmed by several findings of paleoanthropologists, showing that the mechanical dexterity of the human hand has been a major factor in allowing homo sapiens to develop a superior brain [2].

As functional extension of human hand, robotic hand would undertake more accurate, sophisticated and tedious operations in certain areas. While robotic hand grasping planning is a difficult problem in robotics that has occupied researchers since the 1980s. Integrating multisensor data, extracting and modeling these data could greatly alleviate constraints facing in this area. During the past decades, many efforts have been made on how to perform an accurate and stable grasping with the help of different kinds of sensor, e.g. in [3]-[6] visual sensing has been widely used, and tactile sensing are used in [7]-[8]. While as a low-cost 3D camera, *Microsoft Kinect*TM, sampling color image and depth image, has not been fully studied and received higher priority in the robotics domain.

In this paper, we are trying to use tactile data sampled from sensors (see Fig. 1) mounted on the surface of each

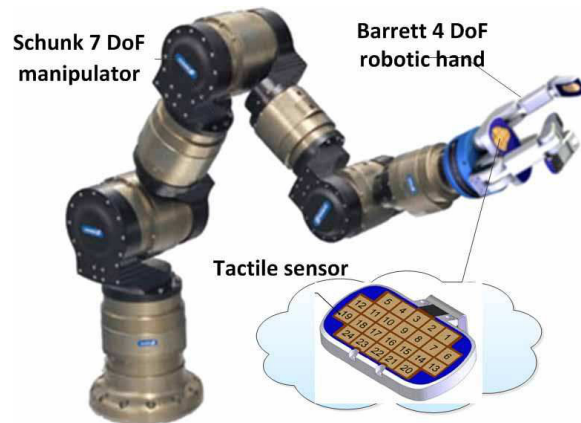


Fig. 1. Dexterous robotic hand grasp system with 7 DOF *Schunk*TM manipulator and 4 DOF *BarrettHand*TM robotic hand.

finger of robotic hand, with the help of depth data from *Kinect*TM deployed beside target objects, accomplishing target stable grasping coordinately, using our 7 DOF *Schunk*TM manipulator and 4 DOF *BarrettHand*TM robotic hand. Under the guidance of *Kinect*TM, manipulator is moving close to the target, and when fully getting into the workspace of dexterous hand, *BarrettHand*TM performs tentative grasping, in the meantime, stable grasping hand pose learning proceeds using the proposed model. After several off-line training, one of best hand pose could be selected and thereafter *BarrettHand*TM could perform stable and accurate grasping according to different types of object. So far, it is a preliminary trial using depth information, further research concerning multi-level integration with tactile information would be studied in the near future.

It is important to point that grasping hand pose learning conveys the final stable grasping during the whole grasping phases. Generally, a variety of learning problems in robotic hand grasping can be construed as problems of modeling dynamical process from time sequential observations. Two major approaches for modeling dynamical systems in machine learning are latent variable models (LVMs) and predictive models, which have different benefits and drawbacks [9]. The two of effective instances of LVMs are Hidden Markov Models (HMMs) and Linear Dynamical Systems (LDSs) or Kalman filter model. In this paper, we focus on how to fit the multi-phase grasping and obtain a proper parametric model to better reproduce the whole grasping process. To that end, we present a learning and planning method based the

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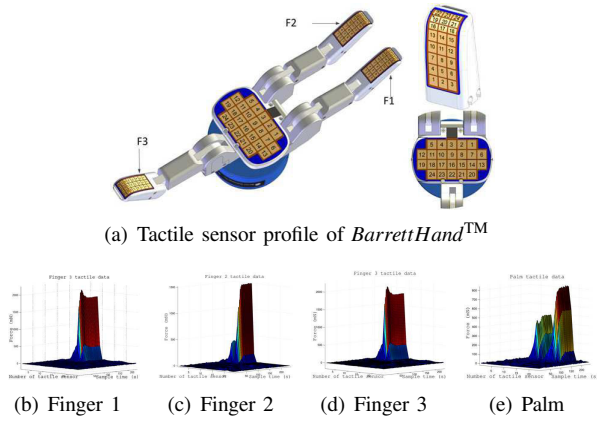


Fig. 2. Tactile sensor profile and distributions of tactile-force

proposed model inspired by LDSs to assess grasp stability and demonstrate that knowledge about grasp stability can be inferred from data of tactile sensor. The contributions of our work and the state-of-the-art are discussed in the Section II. The theoretical framework is detailed in Section III. We present experimental evaluation and analyze the data collected from real target grasping in Section IV. Finally we summarize our work and propose the future work in Section V.

II. RELATED WORK AND OUR CONTRIBUTION

In order to establish a framework of stable grasp hand gesture learning and planning, one of touchy issues confronting us is how to improve our powers of perception on the stable grasping and recognition of the relationship between multisensor data fusion and stable grasping hand pose evaluation mechanism. The rest part will introduce the prior effort on this issue together with our attempt and contribution.

A. Related Work

In the area of robotic object grasping, there has been a lot of effort during the past decades, in [2], Antonio Bicchi summarizes the evolution of robotic hand grasping, with a critical evaluation of what in the author's view are the leading ideas and emerging trends is privileged with respect to exhaustiveness of citations. Springer Handbook of Robotics [10] contains recent survey, which brings a widespread and well-structured compilation of classic and emerging application areas of robotics, especially on robot hand structures, sensing, perception and manipulation. As stated in [11], most of grasp planning approaches first model the target object shape with a number of primitives such as rectangle [12], spheres [13], or super-quadratics [14] to facilitate grasping, hand pose (both position and orientation) and force-exertion strategies are then generated based on analytical grasp quality measures, which rely heavily on accurate knowledge of the contacts between the hand and the target object in order to estimate the stability of a hand pose and corresponding magnitude of force. [15] and [16] report related researches, yet the drawback of this approach they

adopted lies in strongly dependent on the visual information about the target object. For this reason, it is too theoretical and lack of practicality. Distribution and magnitude of tactile data from each finger of robotic hand reflect meticulous state of fingertips and the object, therefore this type of sensor should receive higher priority especially in accurate and stable grasping. [9] and [11] report related researches, yet they put more emphasis on the importance of information from contact region, while ignore important role of visual information modeling shape of the object and helping to orchestrate manipulator approaching the target object.

Besides, there is some work related to object shape learning based on tactile data, such as [17] and [18] using snapshot data from final state of the contact, [7] [8] and [19] using time series tactile data sampled throughout the grasping execution. In [7] and [8], a system that integrates grasp planning and online grasp stability assessment is presented.

In summary, multi-sensor integration such as vision, tactile and other types of sensors would be beneficial to analyzing stability and reliability of robotic grasping, for instance, visual information could cover the shortage of tactile data when manipulator approaching the target objects before contacting, while tactile data could play an important role in reflecting meticulous state of fingertips and the object on the contact surface in turn.

B. Contributions of this Work

In this paper, we take full advantage of tactile sensor (collecting contact force, see Fig. 2) accomplishing accurate and stable grasping. Besides, depth information plays an important role in the dextrous robotic hand grasping process.

As for tactile data time-series analysis, we model the whole process of grasping using the proposed so-called piecewise LDSs, which could better describe the distribution and magnitude of tactile data time-series than LDSs.

III. METHODOLOGY

To facilitate theoretical description, we summarize the whole grasping process in brief: we are trying to use tactile information accomplishing target stable grasping, using our 7 DOF *Schunk*TM manipulator and 4 DOF *BarrettHand*TM. Under the guidance of 3D reconstruction of scenes based on color and depth information sampled by *Kinect*TM, we could operate the manipulator heading to the target, and when getting into the workspace of dextrous hand, *BarrettHand*TM performs tentative grasping, at the same time, stable grasping hand pose learning and planning proceed using the proposed model. After several off-line training, one of best hand pose could be selected and then *BarrettHand*TM performs stable and accurate grasping according to different types of object. To insure stable grasping hand pose learning and selecting, a proper model, accurately describing the changing pattern of tactile information during the whole grasping process, seems to be important. The following part would make an effort on finding this proper model.

We collect the tactile reading during the whole process starting from the moment when we operate the manipulator

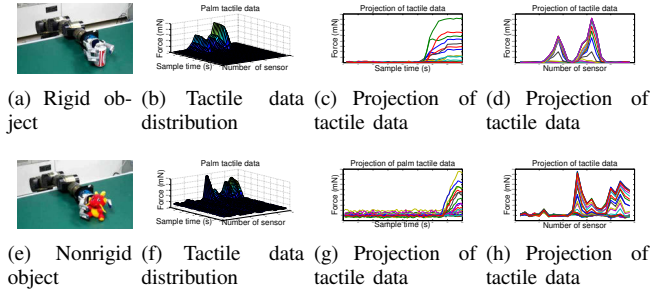


Fig. 3. Rigid and nonrigid object grasping data analysis

approaching the target until the end of the grasp which means the fingers are closed and the joints do not move any more, which is illustrated in Fig. 2. As we can see, tactile data on each part of four tactile sensor arrays mounted on three fingers (F1, F2 and F3) and palm respectively. It is observed that the distribution and magnitude of tactile-force are similar among three fingers; yet the palm presents a different pattern obviously (with two peak value). Fig. 3 gives detailed information about the palm data with projection along different orientations. Why does it happen? It is believed that the palm has broader contact area than any other three fingers (see Fig. 2), as a result, when the target object is wrapped by the entire hand tightly, the palm hold more contact point. This is especially true for nonrigid target object (see Fig. 3). A distinguishing characteristic of tactile data observations (hereafter also called fingerprints) exhibit temporal correlations, modeling of which is the main challenge facing dynamical systems analysis. Accurate models of time-series fingerprints allow us to perform a variety of useful tasks, such as distribution comparison, stability recognition and prediction, gesture of the hand selection and grasp planning. So it is important to obtain an effective model.

A. Typical Model

One of major approaches for modeling these dynamical time-series data is LVMs assuming its observations are noisy emissions from an underlying latent variable that evolves over time and represents the state of the system [9].

Two of useful instances of LVMs for dynamical systems are HMMs and LDSs. Different from HMMs, LDSs assume a continuous latent variable that evolves linearly with Gaussian noise, which makes LDSs suitable for modeling smoothly evolving dynamic. Furthermore, matrix-factorization-based (e.g. singular value decomposition) approaches to learning parameters of LDSs make it easy to perform model approximation. It is noted that conventional learning (EM) algorithms for the LDS do not guarantee stable parameters for modeling its dynamics [9] [20] [21].

As stated above, fingerprints exhibit temporal correlations, which provide more possibilities of using the core ideas of LDSs and representing the real state of the system. Inspired by [21], we are trying to model tactile time-series data, using so-called piecewise LDSs (switched LDSs or multi-LDSs), better “interpreting” and “understanding” data series to further inferring a joint stochastic model who generates

the data, and then find relationship between distribution of fingerprints and grasping stability.

As illustrated in Fig. 3 (c) and (g), fingerprints time-series present apparent variation regularly over time, which could be mainly divided into three phases, (1) “Inception”, during which, manipulator is approaching the target object, and robotic hand does not get into its workspace, so tactile data is mainly composed of zero wandering of sensors and turbulent noise; (2) “Ascent”, this is a transient process, robotic hand become contacting the object and with contacting area increasing, the average value of exerting force is increasing two, until reach threshold set beforehand in order to protect tactile sensor from abnormal contacting. And (3) “Plateau”, which is the other steady state, yet different from inception phase. In plateau phase, robotic hand has already grasped the target object, and due to the force threshold, it does not exert greater force but keep the same magnitude as in the ending of ascent phase.

B. Definition of Dynamic Process

In order to better analyze this dynamic process of fingerprints, before we model this process, variables definition are given below. Let $\{D(t)\}_{t=1 \dots \tau}$, $D(t) \in \mathbb{R}^m$, be a time-sequence tactile τ data without noise, each $D(t)$ is a slice of the time-sequence at time of t . Thus we can get the real measurement $\{y(t)\}_{t=1 \dots \tau}$ of the tactile data with noisy added, i.e. $y(t) = D(t) + w(t)$, where $w(t) \in \mathbb{R}^m$, $t = 1 \dots \tau$. We say that the sequence $\{D(t)\}$ is a (linear) dynamic process of fingerprints if there exists a set of n spatial filters $\phi_\alpha : \mathbb{R} \rightarrow \mathbb{R}^m$, $\alpha = 1 \dots n$ and a stationary distribution $q(\cdot)$ such that, defining $x(t) \in \mathbb{R}^n$ ($m \gg n$) such that $D(t) = \phi(x(t))$, we have $x(t) = \sum_{i=1}^k A_i x(t-i) + Bv(t)$, with $v(t) \in \mathbb{R}^{nv}$ and identically distributed (IID) realization from the density $q(\cdot)$, for a certain way of choosing $A_i \in \mathbb{R}^{n \times n}$, $i = 1, \dots, k$, $B \in \mathbb{R}^{n \times nv}$ and initial condition $x(0) = x_0$. Therefore, a linear dynamic process of fingerprints is a special case of an autoregressive moving average process (ARMA):

$$\begin{cases} x(t+1) = Ax(t) + Bv(t) \\ y(t) = \phi(x(t)) + w(t) \end{cases} \quad (1)$$

Detailed information about system (1) could be found in [21].

Piecewise LDSs comprises a collection of system (1), i.e. every system within the collection has the same form as system (1), together with a switching rule that specifies the switching between the subsystems. Piecewise LDSs have the advantage to accurately cope with complex dynamic process or systems with large uncertainties, especially the process comprise different phases showing obviously different characteristics.

Mathematically, piecewise LDSs can be described by:

$$\begin{cases} x(t+1) = A_\sigma x(t) + v(t) \\ y(t) = C_\sigma x(t) + w(t) \end{cases} \quad (2)$$

with the switching signal:

$$\sigma(t_+) = \sigma(x(t)|y(t), \sigma(t_-)), \quad \forall t \quad (3)$$

where $x(t)$, $v(t)$, and $w(t)$ are the same as previously mentioned, while A_σ and C_σ are of a collection of $\{A_i\}$ and $\{C_i\}$, and system (2) can be simply denoted as $\Sigma(A_i, C_i)_{\mathcal{J}}$. Matrix pair (A_k, C_k) for $k \in \mathcal{J}$ is referred to as one of subsystems of (2). $\sigma(t_+)$ only depends on its past value $\sigma(t_-)$, latent state $x(t)$ and observation $y(t)$. If $\sigma(t) = i$, we would say that the i^{th} subsystem is *active* at time t . It is clear that at any instant, there is one (and only one) active subsystem.

The performance of the overall system relies on every subsystem and the switching series. Different switching signals may incur totally different system behaviors. Here we adopt time-driven switching law to model the dynamic fingerprints called piecewise LDSs with time-driven switching law. Within every time slot, our dynamic fingerprints take on different types of system characteristic respectively. As stated in typical model subsection, we have divided the time-series into three phase: (1) Inception, (2) Ascent and (3) Plateau, hence the corresponding time-driven switching law should be:

$$\sigma = \begin{cases} 1 & t \in \text{Inception} \\ 2 & t \in \text{Ascent} \\ 3 & t \in \text{Plateau} \end{cases} \quad (4)$$

C. Problem Formulation

As shown in Fig. 3, we find that the model (2) captures the gist of distribution and magnitude of tactile data. To be specific, the whole process is divided into three phase, while every phase is an ARMA dynamic process. Different choice of filters ϕ could induce different types of ARMA dynamic process. A model, whether accurate or not, depends mainly on whether it captures the significant information of a dynamic process. The rest part, which is partly inspired by [21] and its successful application in dynamic texture, would be emphasized on how to deal with every phase (“Inception”, “Ascent” and “Plateau”) and its application in robotic hand grasping.

Taking single phase into account, say, “Plateau”, we adopt linear filters ϕ to reconstruct observations in the preliminary study phase, see (5). This means every $D(t)$ is linear combination of $x_i(t)$, $i \in 1 \dots n$, with combination coefficient $\{\theta_i\}$. As for how to specific parameters, it is common to select Maximum Likelihood Estimation(MLE) for its convenient and efficient. By decomposing the fingerprints $D(t)$, each column of C could be orthogonal, i.e. $\{\theta_i\}$ is orthogonal:

$$D(t) = \sum_{i=1}^n x_i(t) \theta_i = Cx(t) \quad (5)$$

where $\{C\} = [\theta_1, \dots, \theta_n] \in \mathbb{R}^{m \times n}$.

Therefore, the problem of modeling the fingerprints has been transformed into parameters estimation of A , B , C , q , given $y(1), \dots, y(\tau)$:

$$\begin{aligned} \hat{A}, \hat{B}, \hat{C}, \hat{q}(\cdot) &= \arg \max_{A, B, C, q} \log p(y(1), \dots, y(\tau)) \\ \text{s.t. } Eq.(1), v(t) &\stackrel{ID}{\sim} q \end{aligned}$$

Furthermore, we assume there exist a symmetric positive-definite matrices $Q \in \mathbb{R}^{n \times n}$ and $R \in \mathbb{R}^{m \times m}$ such that

$$\begin{cases} x(t+1) = Ax(t) + v(t) \\ y(t) = Cx(t) + w(t) \end{cases} \quad (6)$$

where $v(t) \sim \mathcal{N}(0, Q)$ and $w(t) \sim \mathcal{N}(0, R)$. Now the problem has been transformed into parameters estimation of A , C , Q , R from the observations $\{y(t)\}$. Note that stochastic disturbance $Bv(t)$ in (1) is slightly different in form to $v(t)$ in (6).

$$BB^T = Q \quad (7)$$

and $v(t) \sim \mathcal{N}(0, I_{nv})$ i.e. $Bv \sim \mathcal{N}(0, Q)$ in (1), yet in (6) $v(t) \sim \mathcal{N}(0, Q)$.

We could further specify the problem of parameters estimation of A , C , Q , R , as follow, given $y(1), \dots, y(\tau)$:

$$\begin{aligned} \hat{A}, \hat{C}, \hat{Q}, \hat{R} &= \lim_{t \rightarrow \infty} \arg \min_{A, C, Q, R} E \|y(t+1) - Cx(t+1|t)\|^2 \\ \text{s.t. } &Eq.(1) \end{aligned}$$

According to the matrix theory, the choice of A , C , Q , R is not unique. Using similitude transformation, we would obtain a family of A with PAP^{-1} , C with CP^{-1} , Q with PQP^T and R with PRP^T (P is similitude transformation matrix). However, we could choose a so called canonical model represent each equivalent one. The canonical model has more convenient and efficient for analyzation, numeration, simulation and prediction of dynamical system behavior. So we choose the model that makes the columns of C orthogonal, $rank(C) = n$. These assumptions could meet easily by truncated singular value decomposition(TSVD), which provide a effective way of solving the matrix low rank approximation to canonical matrix. Detailed description will be given bellow.

D. Closed-Form Analytic Solution

Rewriting latent state $x(t)$ and observation $y(t)$ into matrix form, let $X_1^\tau \doteq [x(1), \dots, x(\tau)] \in \mathbb{R}^{n \times \tau}$, $Y_1^\tau \doteq [y(1), \dots, y(\tau)] \in \mathbb{R}^{m \times \tau}$ and $V_1^\tau \doteq [v(1), \dots, v(\tau)] \in \mathbb{R}^{n \times \tau}$, $W_1^\tau \doteq [w(1), \dots, w(\tau)] \in \mathbb{R}^{m \times \tau}$, and by (6), we have matrix form:

$$\begin{cases} X_2^{\tau+1} = AX_1^\tau + V_1^\tau \\ Y_1^\tau = CX_1^\tau + W_1^\tau \end{cases} \quad (8)$$

Using TSVD, Y_1^τ could be decomposed into three parts: $Y_1^\tau = U\Sigma V^T$, where $U \in \mathbb{R}^{m \times n}$ and $V \in \mathbb{R}^{\tau \times n}$ are orthogonal matrix, $UU^T = I$ and $V^TV = I$, yet $\Sigma \in \mathbb{R}^{n \times n}$ is diagonal matrix, with $\Sigma = \text{diag}\{\sigma_1, \dots, \sigma_n\}$, σ_i denotes i^{th} singular value of Σ . So the unique estimated value of C and X is given by:

$$\hat{C} = U \quad (9)$$

$$\hat{X} = \Sigma V^T \quad (10)$$

According to $X_1^\tau = AX_1^\tau + V_1^\tau$ in (8), the unique estimated A could be inferred:

$$\hat{A} = \hat{X} \hat{X}^\dagger \quad (11)$$

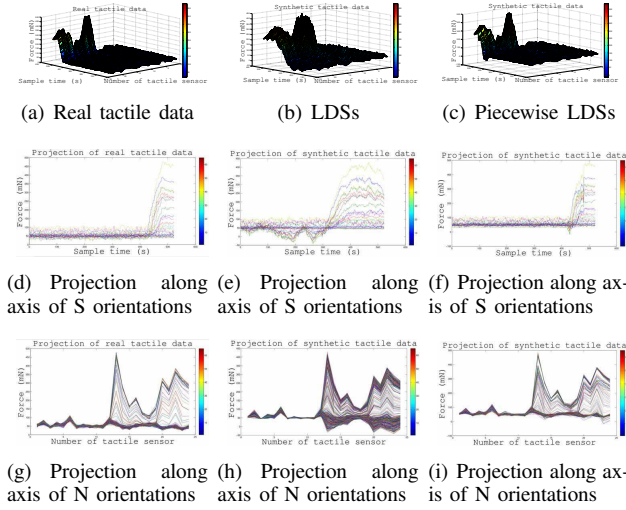


Fig. 4. Distributions of real tactile data vs. synthetic tactile data using LDSs and piecewise LDSs. (S: Sample time; N: Number of tactile sensor)

where $\hat{X}^\dagger$ is the Moore-Penrose pseudoinverse of $\hat{X}$.

Process noise covariance Q could be estimated from V_1^τ :

$$\hat{Q} = E(v(t)v^T(t)) = \frac{1}{\tau-1} \sum_{t=1}^{\tau-1} v(t)v^T(t) \quad (12)$$

Every $v(t)$ is the column of V_1^τ and (12) means the estimated value of Q is weighted average of each covariance of $v(t)v^T$. We can also infer B from (7) by TSVD of V_1^τ or eigenvalue decomposition (EVD) of $\hat{Q}$.

Now, we obtain closed-form analytic solution for single phase. With corresponding switching law (4), we could further model each phase, and find final form of piecewise LDSs.

E. LDSs vs. Piecewise LDSs

Both LDSs and piecewise LDSs are practical in modeling time-series data. Therefore, we have tried these two types of model on our experimental platform. As illustrated in Fig. 4, LDSs show less accurate than piecewise LDSs on tactile data analysis, especially when subsystems comprising entire dynamic process differ widely from each other. Detailed information on performance comparison between these two types of models would be presented in Section IV.

IV. EXPERIMENTAL EVALUATION

To testify the feasibility and accuracy of piecewise LDSs, plenty of experiments have been done on our experimental platform, described before: 7 DOF *Schunk*TM manipulator and 4 DOF *BarrettHand*TM with tactile sensors. Fig. 4 (d) and (g) give us a detailed information on real tactile sensor data. It is observed that in the inception phase ($t = 1 \sim 420s$), observations appear fluctuating between 0 and 100 $m \cdot N$, which are mainly composed of noise and stochastic disturbance. However, tactile data in the ascent phase ($t = 421 \sim 480s$) increase rapidly, mainly because in this phase, *BarrettHand*TM is getting into touching the target object and keeping exerting force on the surface. When

reaching threshold of the tactile sensors, we are arriving at a equilibrium, the plateau phase ($t = 481 \sim 522s$), which evolves smoothly and steadily, occasionally disturbed by background noise.

By MLE and TSVD, we could use Y_1^τ training A , C and Q of LDSs and piecewise LDSs respectively, and then generate new time-series of fingerprints $\hat{Y}_1^\tau$ using (8) for LDSs, (2) and (4) for piecewise LDSs. We could observe that, (see Fig. 4 (b) and (c), synthetic data generated by LDSs and piecewise LDSs respectively) LDSs does not fully describe a whole process of tactile although its projection along axis of N orientation, (see Fig. 4 (h)), seems like its real data counterpart, (see Fig. 4 (g)), while piecewise LDSs provide relatively nice product modeling the same process, see Fig. 4 (i). More rigorous performance comparisons are given in Table I, where $e = Y_1^\tau - \hat{Y}_1^\tau$, denotes error between real data Y_1^τ and synthetic data $\hat{Y}_1^\tau$. It is demonstrated that piecewise LDSs show a good agreement with an error less than 10%.

TABLE I

LDSs VS. PIECEWISE LDSs SYNTHETIC PERFORMANCE COMPARISON

	Real data	LDSs	Piecewise LDSs
$\ Y_1^\tau\ $	5595.21	5595.22	5595.21
$\ \hat{Y}_1^\tau\ $	—	11192.10	9092.22
$\ e\ $	—	6092.91	619.94
$\ e\ /\ \hat{Y}_1^\tau\ $	—	54.44%	6.82%

Note: $\|\cdot\|$ denotes l_2 -norm of matrix. — denotes not-exist.

Table II demonstrates piecewise LDSs synthetic performance during every phase. And it is noted that overall performance is relative nice, with 6.82% average model error, though 12.48% peak value during ascent phase. The reason is we adopt linear model to approximate the phase, yet higher order terms are truncated, which then incur inaccuracy. It is a practicable approach in our preliminary study, and subsequent improvement will continue. In order to show

TABLE II

PIECEWISE LDSs SYNTHETIC PERFORMANCE DURING EVERY PHASE

	Inception	Ascent	Plateau	Whole
$\ e\ $	280.90	580.18	152.32	619.94
$\ \hat{Y}_1^\tau\ $	5606.72	4648.7	6340.13	9092.22
$\ Y_1^\tau\ $	5595.23	4968.01	6243.62	9178.31
$\ e\ /\ \hat{Y}_1^\tau\ $	5.01%	12.48%	2.40%	6.82%
$\ e\ /\ Y_1^\tau\ $	5.02%	11.68%	2.44%	6.75%

the merits of the piecewise LDSs, we run both LDSs and piecewise LDSs for one hundred times respectively, and illustrate the comparison curves, shown in Fig. 5 (Rigid object grasping) and Fig. 6 (Nonrigid object grasping). Mathematical expectations(mean) and variances of different types of grasping tactile time-series are shown in Table III. Corresponding to Fig. 5, Table III gives detailed mathematical comparison. We can obviously observe that mean value of relative error $\|e\|/\|Y_1^\tau\|$ of piecewise LDSs is reduced, in the

TABLE III

RIGID OBJECT GRASPING SYNTHETIC PERFORMANCE COMPARISON

	LDSs	Piecewise LDSs
Mean of $\ e\ /\ Y_1^r\ $ (rigid)	0.8560	0.2522
Variance of $\ e\ /\ Y_1^r\ $ (rigid)	0.0368	0.0022
Mean of $\ e\ /\ Y_1^r\ $ (nonrigid)	0.6682	0.2159
Variance of $\ e\ /\ Y_1^r\ $ (nonrigid)	0.0892	0.0071

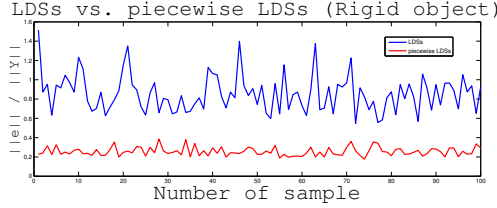


Fig. 5. Rigid object grasping performance comparison

meantime, its variance is reduced dramatically. Fig. 6 shows the same feature in the case of nonrigid object grasping.

In summary, in a sense, piecewise LDSs performs better than LDSs in modeling tactile time-series during robotic hand grasping, with higher accuracy and lower variance, which enable us to better analyze the whole grasping process and select a proper hand pose whose fingerprints will better accord with the stable one.

V. CONCLUSIONS AND FUTURE WORK

In this paper, we propose a novel piecewise LDSs model for robot grasping system in unknown environment via tactile sensors. The model is inspired by LDSs and switched systems in order to better represent and “understand” the whole dynamic process of tactile time-series. It is observed that the “Plateau” phase plays an important role in grasp learning and planning. Every grasping gesture would be mapped into a special type of “Plateau”, therefore, a corresponding grasping gesture could be selected. Detailed mechanism need further be studied. Thus in our future work, we will focus on more accurately modeling the whole process of tactile time-series incorporated with other types of sensors in order to better select a proper gesture and perform a stable grasping.

ACKNOWLEDGMENT

This work was jointly supported by National Natural Science Foundation of China (Grants No. 90820304, 61075027)

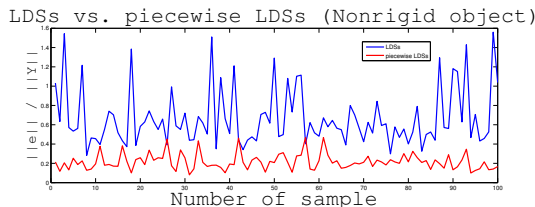


Fig. 6. Nonrigid object grasping performance comparison

and the Tsinghua National Laboratory for Information Science and Technology (TNList) Cross-discipline Foundation (Grant 042003023). The authors gratefully acknowledge the financial support.

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